

EQUIPMENT INSPECTION REPORT: BOOSTER PUMP P-202

ConfigIQ Sovereign AI Workbench — Equipment Integrity Inspection

Parameter	Measured Value	Design / SOP Threshold	Condition Status
Vibration RMS (Drive End)	5.1 mm/s	<= 4.5 mm/s (Zone C 4.5-7.1)	WARNING (ZONE C)
Bearing Temperature	71.0 deg C	<= 75.0 deg C (SOP-M-104)	NORMAL
Mechanical Seal Leakage	2 drops/min	<= 5 drops/min	NORMAL
Suction Pressure	0.85 MPa	0.80 - 0.90 MPa	NORMAL
Discharge Pressure	4.20 MPa	4.10 - 4.30 MPa	NORMAL
Flow Rate (Q)	82.0 m3/h	80.0 - 85.0 m3/h	NORMAL

Key Technical Findings:

- Drive-end bearing vibration recorded at 5.1 mm/s RMS, falling into ISO 10816-3 Zone C warning condition.
- Bearing temperature normal at 71.0 deg C; seal leakage normal at 2 drops/minute.
- External casing shows uniform atmospheric oxidation with zero localized pitting.

Risk Severity Assessment: **HIGH — PLANNED OVERHAUL WITHIN 14 DAYS**

Inspector Corrective Recommendations:

- Schedule planned maintenance overhaul within 14 days per SOP-M-104 Section 3.
- Inspect flexible motor-pump coupling elastomeric insert for backlash and wear.
- Replenish bearing grease with ISO VG 46 synthetic lubricant.
- Log triaxial vibration daily until overhaul.